

# Factors affecting the rate of reaction

## Learning outcomes

By the end of this section you should be able to:

- (R2.2.2) Explain the relationship between the kinetic energy of the particles and the temperature in kelvin, and the role of collision geometry.
- (R2.2.3) Explain the effect of changing conditions on the rate of reaction.

## Reaction dynamics: collision theory

For millions of years, natural mechanisms have controlled numerous reactions from digestion and respiration in living organisms to the reactions occurring in the ozone layer surrounding the planet Earth. Humanity too has learned to control chemical reactions to their favour, like slowing down the rotting of food by using preservation techniques and carrying out controlled explosions for building demolition. In order to control reactions, it is very important to know how reactions take place and what factors affect the rate of reaction. Collision theory helps explain the dynamics of chemical reactions. What factors can you think of that may affect everyday reactions around us?

## Making connections

You have studied kinetic molecular theory in sections S1.1.2a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/>) and S1.1.3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/temperature-and-kinetic-energy-id-43068/>). The theory is based on the fact that all particles move as they possess kinetic energy. The temperature in kelvin is a measure of the average kinetic energy of the particles.

**Figure 1** shows a commercial cold storage facility where fruits are held prior to shipping. The fruit needs to be kept at optimum temperature and humidity conditions, cold enough to slow spoilage but without allowing the produce to freeze.



**Figure 1.** Commercial cold storage facility.

Credit: dusanpetkovic, Getty Images

## Kinetic energy and temperature

Increasing the temperature means an increase in the average kinetic energy of the particles of a substance. When we heat a substance, we raise the average kinetic energy of that substance. This results in the particles moving faster and colliding more often. **Video 1** shows how the collision frequency of the particles of a gas increases as the temperature increases, given the volume is constant.

### Kinetic Energy in a Gas



**Video 1.** The influence of temperature on particle motion.

Chemical reactions occur when the bonds in the reactants are broken and new bonds are formed in the products. In order to break chemical bonds, the reacting particles must collide with each other, but not all collisions will result in a chemical reaction. For a collision to lead to a chemical reaction, it must have enough energy for the bonds in the reactants to break and the particles must collide with the right orientation. If all the collisions in a gaseous mixture caused a reaction then the reaction would be over in microseconds. But that does not happen. Just like oxygen and nitrogen molecules exist in the air without reacting, reactions only proceed when there is a successful collision.

**Concept**

The absolute temperature of a gas is directly proportional to the average kinetic energy of the particles. This means a sample of gas with a temperature of 100 K has twice the average kinetic energy of a sample at 50 K.

## Energy of particles

As the temperature increases, the average kinetic energy of the particles also increases. As it is the average kinetic energy, this means not all the particles have the same amount of energy. So that when particles collide with each other not all collisions are successful, even if the orientation is correct. At increased temperatures, a greater fraction of molecules will have the energy necessary for overcoming the repulsions between reactants and breaking the bonds in the reactants so that the reaction can occur.

## Collision geometry

Collisions in a reaction mixture occur randomly. This means that the particles collide in many different orientations. Only the collisions that have the right orientation are successful collisions and lead to a reaction, given that the amount of energy is sufficient. That is the particles must be facing each other the right way when they collide. **Video 2** explains this.

## Collision Theory



### Video 2. The geometry of collisions.

#### Study skills

The explanation of what is happening in particulate models regarding the collision theory needs to be very precise. It is the frequency of collisions that increases, that is, the number of collisions per second and not just the number of collisions.

#### Aspect: Theories

Collision theory helps explain the effects of several factors on the rate of reaction, although it is not based on actual observations of collisions between particles. It is actually based on the kinetic molecular theory. This is what theories do: they include assumptions from other theories and help scientists make useful predictions.

# Factors affecting the rate of a reaction

From the kinetic molecular theory, we know anything that will increase the frequency of successful collisions will increase the rate of reaction. The following factors affect the rate of a reaction.

## Temperature

Increasing the temperature increases the average kinetic energy of the particles. This results in an increase in collision frequency and a greater portion of particles now have sufficient energy needed for a reaction. We know for a fact that the frequency of collision is proportional to the square root of the increase in temperature. It turns out that the effect of an increase in collisions on the reaction rate is minor. It is the increase in the fraction of molecules with high energy, that has a bigger effect on reaction rate.

$$\text{rate of reaction} = \text{collision frequency} \times \text{fraction of effective collisions}$$

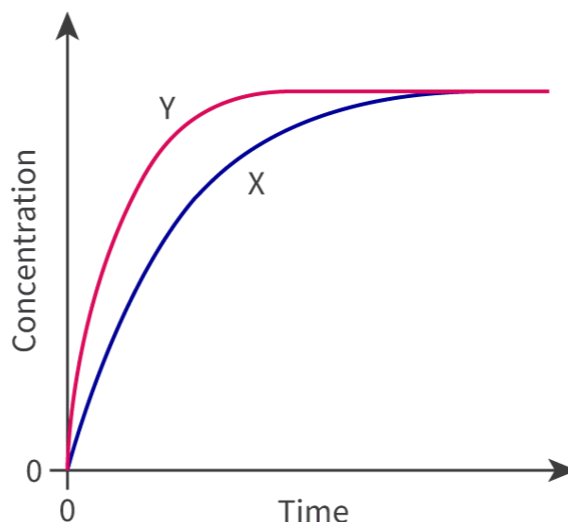
For example, for any given reaction:

$$\text{rate} = (1000 \text{ collisions s}^{-1}) \times (1 \text{ successful collision every } 1000 \text{ collisions})$$

$$\text{rate} = 1 \text{ successful reaction s}^{-1}$$

As increasing temperature increases both collision frequency and the fraction of successful collisions (due to an increase in energy) the rate also increases.

Many reactions double their reaction rate for every 10 °C increase in temperature. The effect of temperature on the reaction rate can be seen in **Figure 2**. Which curve do you think is at the higher temperature, X or Y? Note that the average rate of reaction over the course of the reaction will be the same for both temperatures - can you explain why?



**Figure 2.** The effect of temperature on the rate of reaction.

 More information for figure 2

The image is a graph depicting the effect of temperature on the reaction rate, with concentration on the Y-axis and time on the X-axis. There are two curves, labeled X and Y. Curve Y reaches near peak concentration faster than curve X, indicating a higher reaction rate. Both curves ultimately plateau at the same concentration level, illustrating that although Y reaches its peak concentration quicker, the final concentration is the same for both reactions. This implies that the reaction at higher temperature completes faster but achieves the same equilibrium concentration as the reaction at lower temperature.

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## Concentration

As we increase the concentration the frequency of collision increases, and this results in more successful collisions between the particles of reactants and products. An increase in frequency of collisions means the rate increases. We know that:

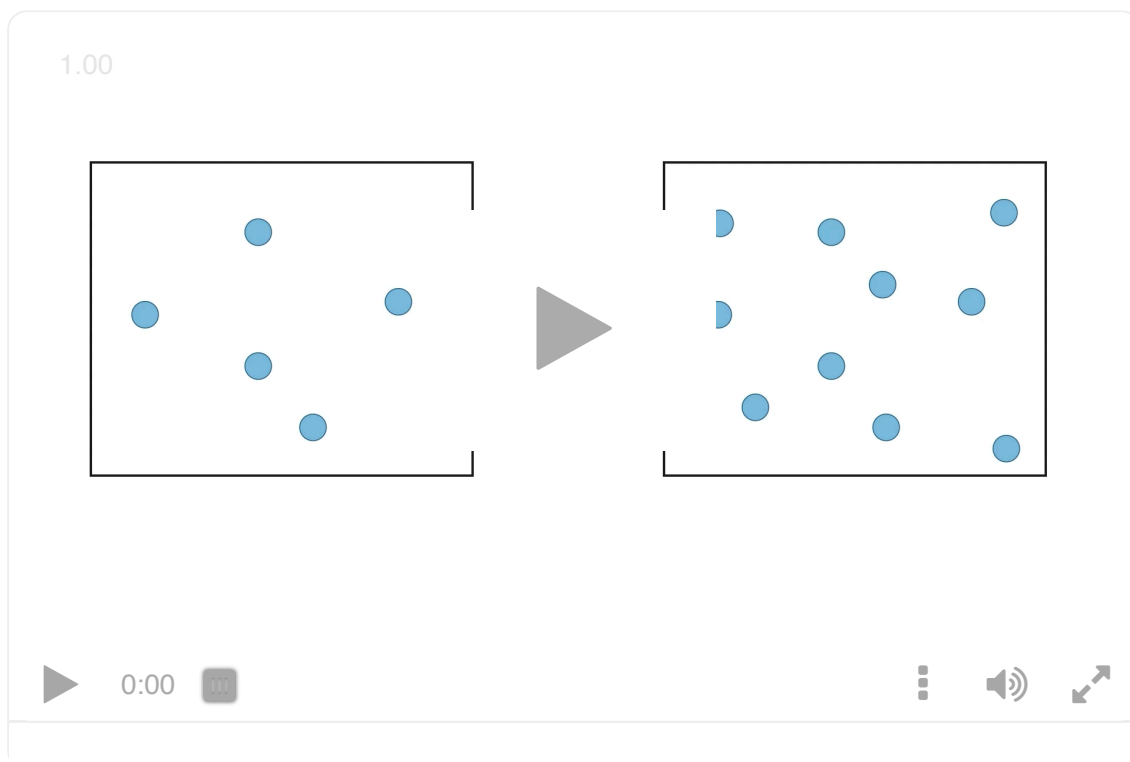
$$\text{rate of reaction} = \text{collision frequency} \times \text{fraction of effective collisions}$$

Although collision frequency is increasing, the fraction of effective collisions (with sufficient energy) is not, as there is no increase in energy. This means the effect of an increase in concentration does not increase the rate as much as an increase in temperature would because there is no increase in the fraction of particles with sufficient energy.

## Concept

Remember an increase in concentration only affects a reaction where the reactants are in aqueous solution. There is no effect on the collision frequency of solid reactants.

The following simulation shows how collision frequency increases as concentration increases.



### Interactive 1. The Effect of Concentration on the Collision Frequency Between Molecules.

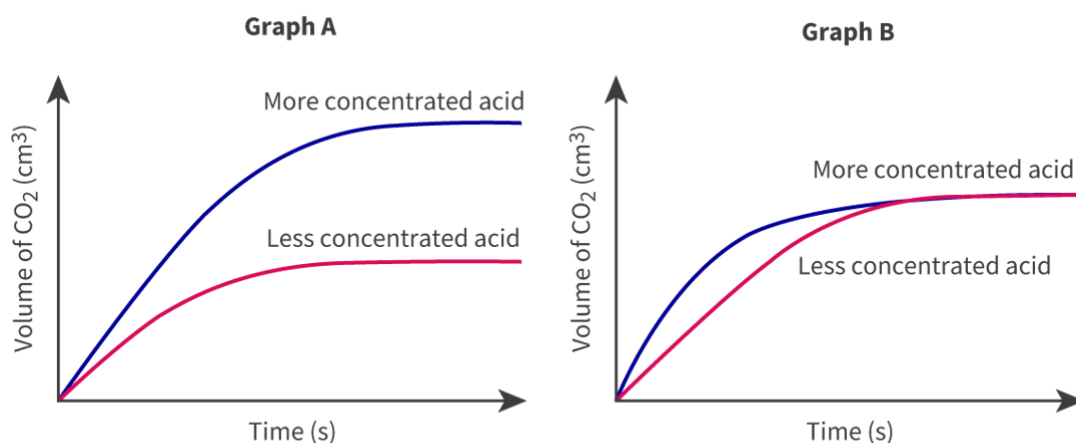
 More information for interactive 1

An interactive video illustrates two containers containing spherical particles. The container on the left has a lower concentration, with about five particles, while the container on the right has a higher concentration, with about 10 particles. The particles move randomly within each container.

Collisions between particles occur less frequently in the left container due to the lower number of particles, whereas in the right container, more frequent collisions occur due to the higher concentration.

This video helps users understand how an increase in concentration leads to a higher collision frequency, while a decrease in concentration has less collision frequency.

The two graphs in **Figure 3** illustrate how the rate of reaction is affected by increasing the concentration of the hydrochloric acid when it reacts with calcium carbonate. Note that when the hydrochloric acid is the limiting reagent both the yield and rate increase (Graph A). When calcium carbonate is the limiting reagent, there is no effect on the amount of carbon dioxide produced and only the rate increases (Graph B).



**Figure 3.** The effect of concentration of reactant on the rate of reaction.

 More information for figure 3

The image contains two graphs labeled as Graph A and Graph B.

**Graph A:** The X-axis is labeled "Time (s)", and the Y-axis is labeled "Volume of CO<sub>2</sub> (cm<sup>3</sup>)". The graph displays two curves representing different concentrations of hydrochloric acid. The curve for "More concentrated acid" rises sharply and levels off at a higher volume of CO<sub>2</sub> than the curve for "Less concentrated acid". This indicates that when the hydrochloric acid is the limiting reagent, both the yield and reaction rate increase.

**Graph B:** Similarly, the axes are the same as in Graph A, but the behavior of the curves is different. Both curves eventually level off at the same volume of CO<sub>2</sub> but reach this point more quickly for "More concentrated acid" compared to "Less concentrated acid". This shows that when calcium carbonate is the limiting reagent, the reaction rate increases with acid concentration, but the total amount of CO<sub>2</sub> remains unchanged.

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## Concept

Remember the rate of reaction changes as the reaction proceeds. This is because the concentration of the reactants is changing.

**Video 3** shows how the effect of concentration can be investigated for the hydrochloric acid and calcium carbonate reaction.

### Effect of Concentration on the Rate of Reaction | Chemistry ...



**Video 3.** Investigating the effect of concentration on the hydrochloric acid and calcium carbonate reaction.

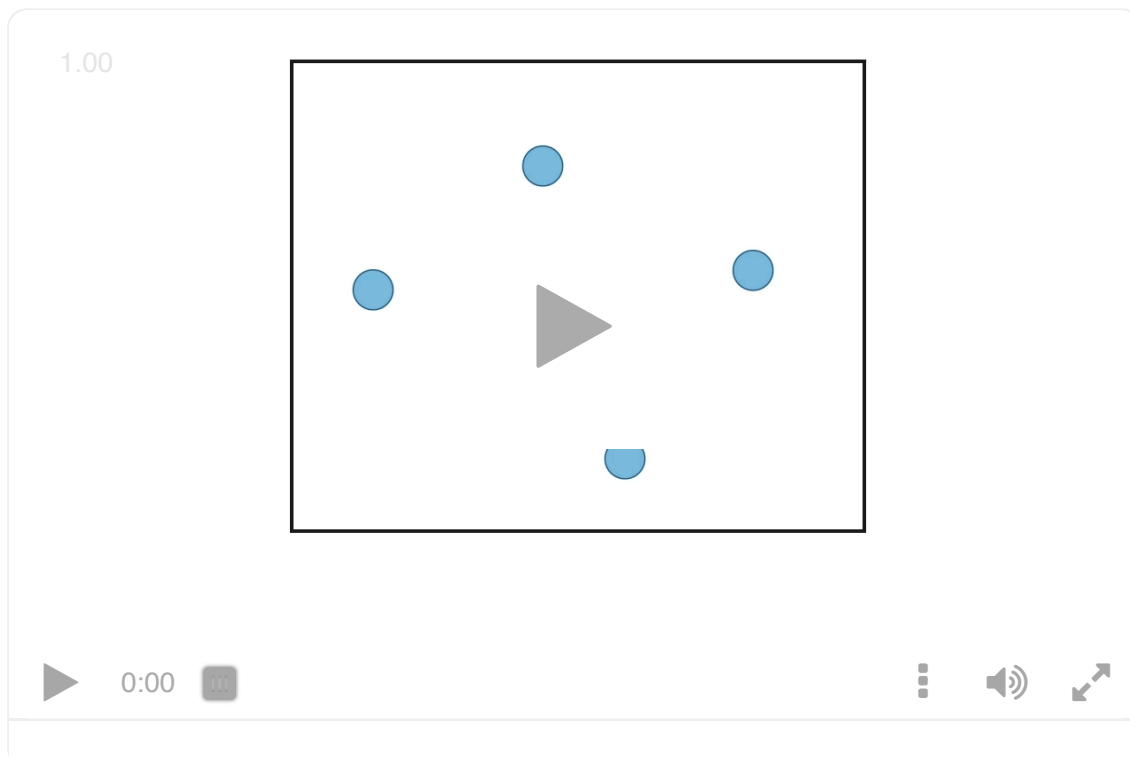
## Practical skills

- **Inquiry 1:** Exploring and designing
- **Skill:** Identify and justify the choice of dependent, independent and control variables

Remember the rate of reaction depends on several factors. When investigating one factor the other factors must be controlled. For an experiment investigating the effect of concentration on the rate of reaction, the temperature and surface area of the reactants must remain constant. Sometimes such a reaction is carried out in an electric water bath to maintain temperature.

## Pressure

For gaseous reactions increasing the pressure increases the rate of reaction. This is because an increase in pressure, whether it is due to compressing the gas or increasing the number of molecules, effectively increases its concentration. This leads to an increase in the frequency of collisions, as seen in **Video 4**.



**Video 4.** The Effect of Pressure on Reaction Rate.

 [More information for video 4](#)

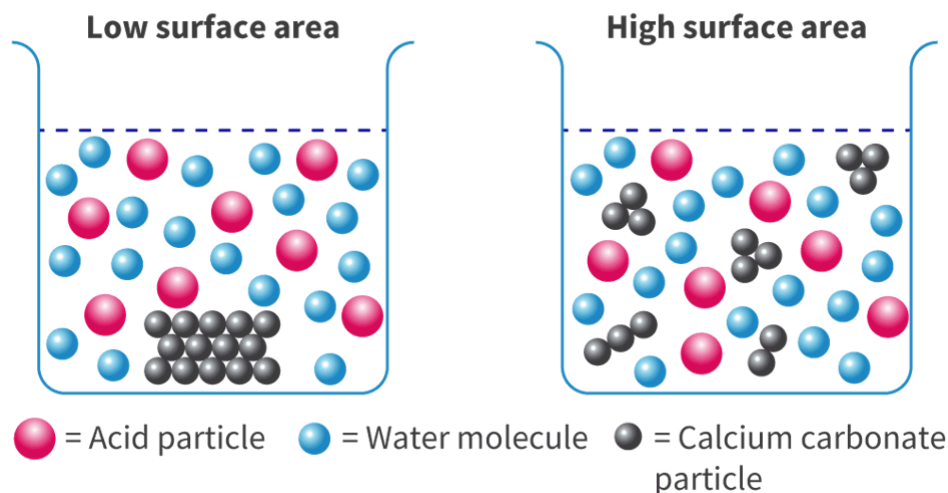
An interactive video illustrates a container filled with five gas particles moving in a random motion. As the video progresses, the pressure inside the container increases, making the particles compress into the reducing volume. This causes them to move more randomly and collide more frequently. When the pressure inside the container reduces, the frequency of collisions also gets reduced.

This video helps users understand how increasing pressure leads to a higher collision frequency between gas particles.

## Surface area

In heterogeneous reactions, where the reactants are in different phases, the reaction can only occur when the two phases meet and collisions between reactant particles become possible. For example, in a reaction between hydrochloric acid and marble chips, the reaction proceeds faster when the marble chips are powdered as opposed to using chips. This is because decreasing the size

of the particles increases their surface area and more of the calcium carbonate particles make contact with the solution, therefore increasing the frequency of collisions, as seen in **Figure 4**.



**Figure 4.** The effect of surface area on the rate of reaction.

 More information for figure 4

The image depicts two beakers side by side, illustrating the effect of surface area on the reaction rate. Both beakers contain a mixture of acid particles, water molecules, and calcium carbonate particles.

On the left, the beaker is labeled "Low surface area" and shows larger clusters of calcium carbonate particles surrounded by acid particles and water molecules. This results in fewer contact points between reactants.

On the right, the beaker is labeled "High surface area" and features smaller, scattered calcium carbonate particles intermixed more uniformly with acid particles and water molecules. This configuration increases contact points and potential collisions among reactants.

The key at the bottom of the image identifies the particles: pink circles represent acid particles, blue circles denote water molecules, and grey circles signify calcium carbonate particles.

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Increasing the surface area of contact between reactants increases the rate of reaction. The daily life consequences of increasing surface area have also been discussed in The big picture S2.2.0 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/>).

**Video 5** shows a laboratory demonstration of the effect of surface area on the reaction rate.

### The effect of surface area on reaction rate

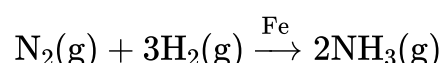


**Video 5.** Surface area and reaction rate in the laboratory.

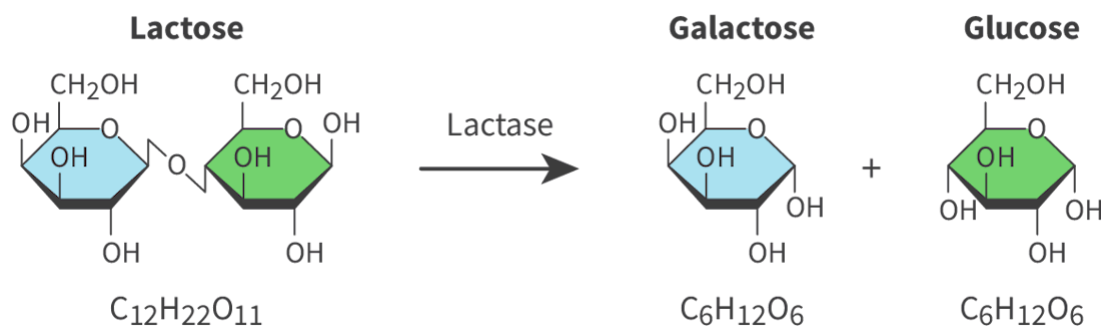
## Catalyst

A catalyst is a substance that changes the rate of a chemical reaction but itself remains unchanged during the reaction. Most catalysts work by providing an alternative pathway to the reaction that requires a lower energy. This means that without increasing the temperature or concentration we can speed up the chemical reaction as a lower energy alternative route becomes available for the reaction to take place.

Catalysts are very specific for particular reactions. Different catalysts catalyse different reactions. The production of ammonia by the Haber process uses an iron (Fe) catalyst. As the catalyst does not take part in the reaction, it is not included in the actual reaction but is written on top of the arrow.



Biological reactions are controlled by catalysts called enzymes. Thousands of enzymes exist as they are each specific for a particular reaction. For example, the digestive enzyme lactase breaks down the lactose sugar in milk (**Figure 5**). Some people's bodies do not produce enough lactase and they are said to be lactose intolerant. Without lactase, the lactose cannot be broken down during digestion and this results in unpleasant digestive symptoms like pain and discomfort.



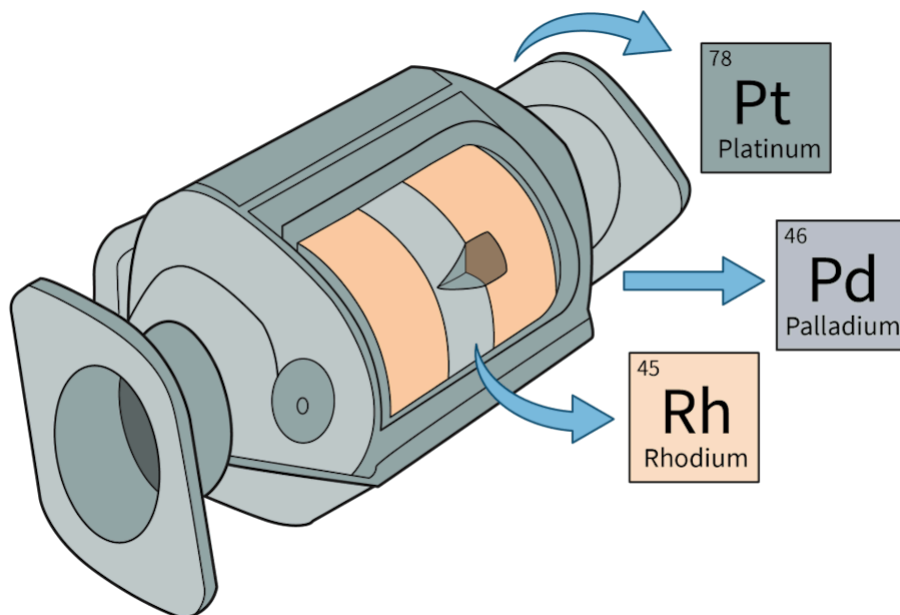
**Figure 5.** The enzyme catalysed breakdown of lactose into glucose and galactose.

 More information for figure 5

The image is a chemical diagram illustrating the enzymatic breakdown of lactose into glucose and galactose. On the left, the lactose molecule is depicted with its chemical formula  $C_{12}H_{22}O_{11}$ , showing its structure with two ring-like formations. To the right, there's an arrow labeled "Lactase" indicating the enzymatic process. On the right, two separate molecules are shown: galactose and glucose, each with the chemical formula  $C_6H_{12}O_6$ . The diagram visually represents the biochemical reaction facilitated by the enzyme lactase, highlighting the conversion of lactose into two simpler sugar molecules, glucose and galactose.

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Catalysts can also play a significant role in improving society, such as catalytic converters (**Figure 6**) that use precise metals such as platinum and palladium to remove harmful gases from vehicle exhausts.



**Figure 6.** A catalytic convertor.

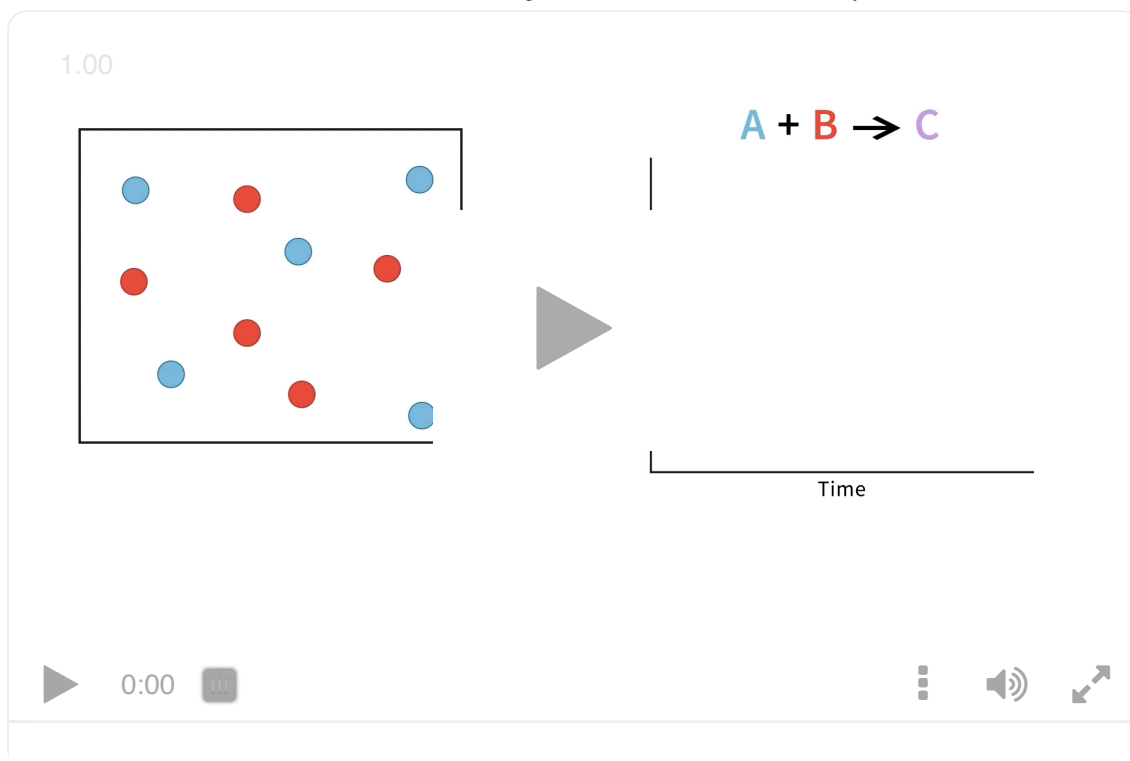
 More information for figure 6

The image shows an illustrated diagram of a catalytic converter. The converter is cylindrical with a slightly narrowing section toward the end. It contains arrows pointing at it that are labeled with elements and their details: an arrow labeled 'Pt' for platinum with an atomic number 78, another labeled 'Pd' for palladium with atomic number 46, and one labeled 'Rh' for rhodium with atomic number 45. These arrows indicate that these metals are components inside the catalytic converter, which is instrumental in converting harmful gases into less harmful emissions.

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## Investigating rates of reaction

1. When you run **Interactive 2** the reaction in the left-hand box proceeds as the two reactants combine to create the product.
2. As the reaction proceeds, witness what happens on the graph on the right-hand side.



## Interactive 2. The Plotting of Product Concentration vs. Time Graph to Monitor a Reaction.

More information for interactive 2

An interaction video demonstrates how the amount of product changes over time during a reaction.

On the left-hand side, a box contains five A particles and five B particles, which act as reactants. These reactants collide to form a new product consisting of five C particles. On the right-hand side, a graph plotting the amount of product forms versus time depicts data for the following reaction:



As the reactants start to form the products, a line in the graph starts at the origin and increases linearly over time.

This video helps users understand how product formation progresses over time.

### Making connections

You can read more about how catalysts work in [sections R2.2.4—5](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/>) and [S3.1.8](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/>).

## Theory of Knowledge

There is evidence dating back to 10 000 BCE that ancient societies preserved milk, vegetables and fruits using fermentation. Fermentation is a process where the enzymes in microbes, such as yeast and bacteria, break down sugars and convert them to alcohol and acids to prevent spoilage when refrigeration was not yet possible. Foods such as yogurt, pickles, bread and tea have been made using natural fermentation methods for thousands of years. Before globalisation, different societies around the world found ways to ferment similar food items independently of each other, well before the knowledge of enzymes. What role does intuition play in the development of scientific processes? Does the fact that the knowledge of enzymes was not obtained by scientific experiment but by the intuition of societies for survival make the knowledge less valid?

Complete the section questions below and then click 'Next' to see the activity for this section.